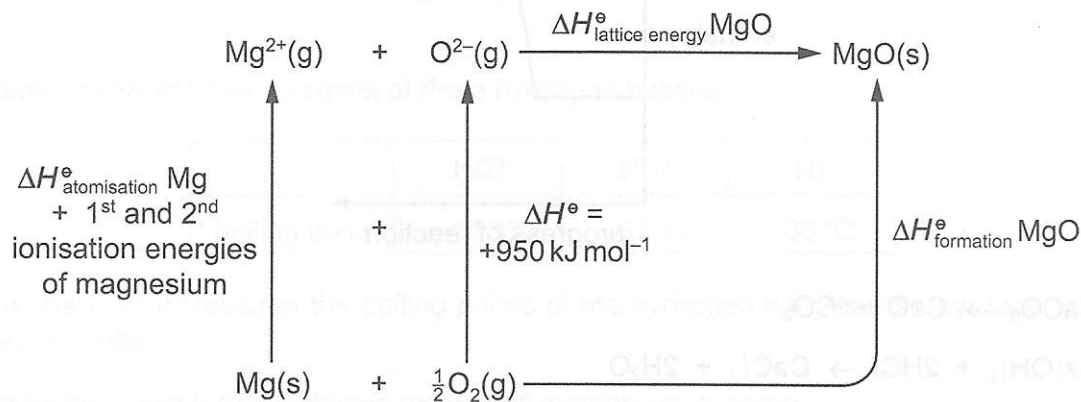


30 A student calculated the lattice energy for magnesium oxide using the cycle shown.

The enthalpy change of atomisation of magnesium, $\Delta H^\circ_{\text{atomisation}} \text{Mg}$, is the energy needed when 1 mol of gaseous magnesium atoms is formed from 1 mol of solid magnesium under standard conditions.



However, the value calculated by the student for the lattice energy was **less** exothermic than the correct value.

Which errors could have been made in the calculation?

- 1 omitting the enthalpy change of atomisation of magnesium
- 2 omitting the first ionisation energy of magnesium
- 3 using the standard enthalpy change of combustion of magnesium rather than the standard enthalpy change of formation of magnesium oxide